

Nutrition Labels for Aging Eyes: Redesigning for Better Health Decisions

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Abstract—The rising proportion of older adults emphasizes the importance of nutritional awareness, as this demographic increasingly relies on nutrition labels when making purchasing decisions. However, preliminary surveys reveal significant misunderstandings among older consumers regarding the information conveyed by standard nutrition labels, leading to suboptimal product choices. To address this issue, we designed an enhanced nutrition label incorporating extended textual and visual explanations—to aid comprehension. In a user study with 24 older adults, participants interacted with the redesigned label during simulated shopping tasks. Our findings indicate that the enhanced label significantly improved participants' understanding of nutritional information, enabling more informed product selections. Moreover, users reported high satisfaction with the experience. These results suggest that enhanced nutritional labels can effectively support older adults in navigating product information, offering a promising direction for future consumer health literacy initiatives.

Index Terms—Older Adults, Visualization, Nutrition Label

I. INTRODUCTION

As the global aging population continues to grow, designing for older adults has become a critical focus in fields like human-computer interaction (HCI) and information visualization. Aging introduces unique challenges, such as declines in visual acuity, cognitive processing, and memory retention, which significantly affect how older adults perceive, process, and interact with information [1]. While visualization tools have been shown to assist older adults in tasks like healthcare decision-making and mobility tracking [2], [3], [4], some everyday activities, such as reading nutrition labels, remain underexplored. Nutrition labels are a vital yet often overlooked source of information for older adults. Unlike other domains, nutrition labels present challenges that even younger users find difficult, such as overwhelming amounts of information, inconsistent layouts, and complex terminology

[5]. Studies suggest that many users, regardless of age, struggle to interpret nutrition labels correctly, often leading to poor dietary choices [6]. However, for older adults, the stakes are higher. Due to their increased vulnerability to chronic conditions like diabetes, hypertension, and cardiovascular diseases, understanding and acting on nutrition label information is crucial for making informed dietary decisions that can directly impact their health [7].

Despite the importance of nutrition labels, current designs fail to address the specific needs of older adults adequately. Research in visualization design for older adults has emphasized simplicity, high contrast, and progressive information disclosure [2], [8], [9]. However, these findings have not been systematically applied to the context of nutrition labels, leaving a significant gap in the design of tools that could empower older adults in their dietary choices. To address this gap, our study focuses on creating and evaluating nutrition label designs tailored to the cognitive and perceptual needs of older adults. We begin by conducting a survey to identify older adults' challenges, preferences, and misconceptions regarding existing nutrition labels. Building on established guidelines, we develop enhanced label designs prioritizing simplicity, clarity, and usability. These designs are then tested with older users to assess their effectiveness in improving comprehension, decision-making, and confidence in food selection.

This research makes three key contributions: 1) Identifying Challenges and Needs: We pinpointed specific issues older adults face with current nutrition labels through the survey. 2) Designing a Tailored Solution: We proposed and developed an enhanced nutrition label application based on visualization principles and user needs. 3) Validating Effectiveness: Through comprehensive testing, we demonstrated the solution's ability to improve comprehension, decision-making,

and user satisfaction, offering insights for future design.

II. RELATED WORK

A. Visualization for Older Adults

With the growing global aging population, increasing attention has been directed toward designing for older adults. Salthouse's processing-speed theory [10] suggests that aging leads to a pervasive slowing of information-processing speed, cascading into declines in attention, working memory, language comprehension, and other higher-order cognitive functions. These age-related changes underscore the need for thoughtful design strategies to ensure that information remains accessible, understandable, and actionable for this demographic [4]. Several research has examined visualization strategies to support older adults. For example, color-coded tables significantly improve their comprehension [11], while abstract or highly interactive designs often lead to confusion and reduced usability [9], [12]. Beyond performance outcomes, studies have identified consistent design preferences. Older adults tend to favor simple, uncluttered layouts with high contrast and large fonts [2], [8], [12]. Participatory design research further shows they value calm, aesthetically pleasing visuals and benefit from contextual cues that support comprehension [13].

B. Nutrition Label Design

Nutrition labels are essential tools for supporting consumers in making informed food choices. Research on nutrition label design can be traced back to the 1980s [14]. We reviewed prior studies in this area and identified two widely adopted strategies: **Graphic Symbols** and **Color Coding**. Mohr, Wyse, and Hansen [14] were among the first to embed bar graphs into the %DV (percent daily value) section of traditional back-of-pack labels. By using length-based visual comparisons rather than numeric text alone, this approach aimed to improve readability and facilitate product comparisons. Subsequent studies by Rudd [15], Geiger et al. [16], and Williams [17] further examined graphical representations in nutrition labels, providing empirical evidence of their effectiveness in enhancing consumer understanding and decision-making. Building on this foundation, Young et al. [18] introduced traffic-light color segmentation (green/yellow/red) within bar graphs to represent low, medium, and high %DV ranges. This modification improved users' ability to identify high-risk nutrients and increased the accuracy of healthfulness judgments. More recent designs continue to expand on these validated strategies. Wimer [19], for instance, used color saturation to encode nutrient levels, combining this with a red-to-green traffic-light scheme to strengthen visual intuitiveness. Coelho et al. [20] employed bullet graphs, where a vertical line marks the actual nutrient level in the product, and two horizontal bars with different color intensities indicate the minimum and maximum range for

that nutrient within the product's subcategory and overall category.

Although previous studies have explored various enhancements to nutrition labels, few have specifically addressed the needs of older adults. Our design fills this gap by improving label visualizations to better support elderly consumers. It is grounded in both established nutrition labeling strategies and empirical insights from our user survey with older adults. This work contributes a novel, user-centered approach that advances nutrition label design beyond conventional paradigms toward greater inclusivity and effectiveness.

III. SURVEY

A preliminary questionnaire was designed to gather background information, assess elderly participants' understanding of nutrition labels, explore their food choices based on such labels, and evaluate their preferences for different levels of information in proposed nutrition label designs. Participants were recruited through an online survey distributed via social media platforms, with a specific focus on older people aged 50 years and above. This age range selection is consistent with that used in previous research [21], [22].

Demographic of Participants. A total of 75 participants aged 50 and above (34 females, 41 males) completed the questionnaire. The majority (50.59%) were aged 50–54, with smaller proportions in other age groups: 55–59 (9.41%), 60–64 (12.94%), and $\leq 3.53\%$ in older brackets (65–80+). Educational backgrounds were diverse: 57.33% held a bachelor's degree, followed by high school graduates (28%), junior high school graduates (8%), and master's degree holders (6.67%). Employment statuses varied, with 36% retired, 14.67% in management roles, and 14.67% freelancers, among others. In terms of health history, 45.33% reported no significant medical conditions, while common conditions included hypertension (34.67%), hyperlipidemia (28%), and diabetes (9.33%).

Elderly Use and Understanding of Nutrition Labels. The survey revealed that nutrition labels play a crucial role in elderly consumers' purchasing decisions. A majority (69.33%) referred to labels when buying products for the first time, and 70.67% used them to compare similar items. However, nearly half (46%) found the information too complex, and 30% reported issues with small or unclear text, limiting accessibility and perceived usefulness. Participants also exhibited significant gaps in comprehension. For example, only 49% correctly calculated total energy consumption based on serving size, and just 14% accurately interpreted NRV% values, often confusing them with total nutrient amounts. When comparing two milk products, while 68% identified the higher-energy option, only 53% selected the better choice for individuals with high blood lipids, highlighting limited understanding of health-related dietary guidance. The survey further indicated a lack of basic nutritional knowledge. On questions about

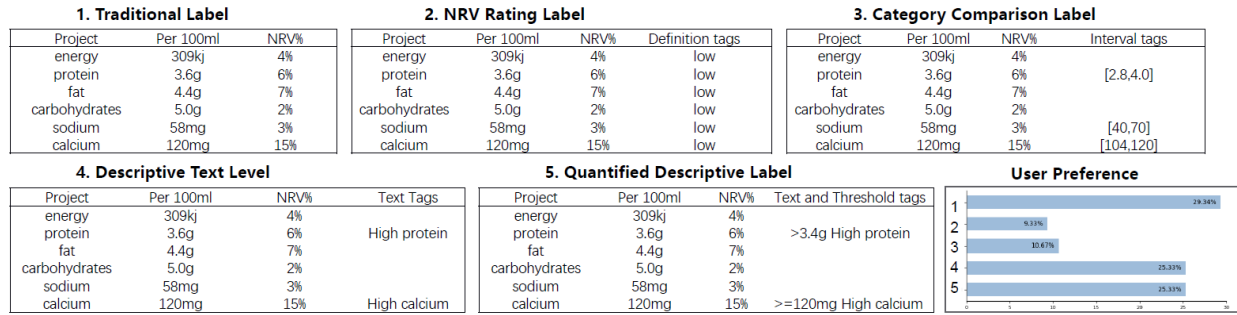


Fig. 1: Five Nutrition label information design & Preferences

trans fats, calcium content in milk, and high-sodium diets, fewer than 50% answered correctly. For instance, only 49% knew that “biscuits, cakes, and pastries” contain the highest trans fats, and just 46% correctly stated that whole and skim milk have comparable calcium levels. Additionally, 49% identified cardiovascular disease as a hazard of high-sodium foods. These findings underline the need for more accessible and educational nutrition label designs tailored to elderly users.

Preferences for Nutrition Label Information Design. To explore the elderly’s preferences for how to interpretate different nutrients in nutrition label, we investigated five presentation methods (Figure 1):

- 1) **Traditional Daily Label:** Displays nutrient values per 100g;
- 2) **NRV Rating Label:** Adds ratings (e.g., “Low,” “Medium,” “High”) based on NRV percentages;
- 3) **Category Comparison Label:** Highlights nutrient ranges within the same food category;
- 4) **Descriptive Text Label:** Provides qualitative descriptions (e.g., “High Calcium”) without numeric details;
- 5) **Quantified Descriptive Label:** Combines qualitative tags with numeric thresholds (e.g., “High Calcium ($\geq 120\text{mg}$)”).

Statistical analysis (Figure 1) showed that 70.67% of participants preferred labels with supplementary information (labels 2–5), compared to the **Traditional Daily Label** (29.33%). Among the enhanced designs, the **Quantified Descriptive Label** and **Descriptive Text Label** received the highest preferences (25.33% each), indicating a strong demand for clear and actionable information. Based on these findings, the **Quantified Descriptive Label** and **Descriptive Text Label** were selected for subsequent prototype design, as they best meet the needs of elderly users for comprehensible and informative labels.

IV. NUTRITION LABEL DESIGN

Based on insights gathered from previous surveys, we discovered that although the majority of elderly consumers read nutrition labels when purchasing food, they encounter significant difficulties (D) in the following areas:

D1: Understanding NRV. Elderly individuals are unaware that Nutrient Reference Values (NRV) are based on a 100-gram serving of the product and do not fully comprehend the meaning of NRV.

D2: Disease-Specific Dietary Needs. They lack knowledge about which nutrients to monitor for specific health conditions.

D3: Impact of Nutrient Levels. They do not understand how the quantity of certain nutrients affects their health.

To address these issues and build upon previous research (reviewed in subsection II-B), we propose a customized solution: an Android application specifically designed for the elderly.

A. Visualization Interface Design

Figure 2 showcases an example of the developed enhanced label application interface. The main design features include seven new design modules aimed at helping elderly users better understand the nutritional components of food and their impacts, providing comprehensive support for health management.

① **Product Total Intake** (Corresponding to **D1**): Displays the specific amount consumed per eating occasion along with corresponding health information.

② **Health Alerts** (Corresponding to **D2**): Highlights specific health warning information within red boxes accompanied by warning icons, alerting specific groups to potential health risks associated with the food.

③-⑤ **Nutrient Information** (Corresponding to **D3**): This comprehensive module encompasses three interrelated components designed to simplify the understanding of key nutrients:

③ **Nutrient Text Tags:** Intuitively labels nutritional components by category, such as “Low Fat” or “High Calcium,” facilitating the identification of key nutrients.

④ **Nutrient Thresholds:** Indicates the defined threshold values for each category, for example, specifying that a product must contain 120mg or more to be classified as “High Calcium”.

⑤ **Nutrient Emoji Tags:** Utilizes intuitive emojis, such



Fig. 2: Enhanced Nutrition Label

as smiling and frowning faces, to represent the health attributes or risks associated with each category, providing a quick visual summary of the nutritional quality.

- ⑥ **Daily Recommended Progress Bar** (Corresponding to **D1**): Shows the percentage of each nutrient in the food relative to the daily recommended intake.
- ⑦ **Nutrient Interpretation** (Corresponding to **D2**): Provides brief explanations of the health benefits or potential harms of each nutrient.

Consistent color coding was applied across the interface, following the established “color coding” strategy introduced in subsection II-B. Red highlights were used for ② Health Alerts, while ③–⑤ Nutrient Information employed green to indicate beneficial levels (e.g., high calcium, low fat, high protein) and red to signal warning levels (e.g., high fat, high sodium). In line with the “graphic symbols” strategy, our nutrition label design also replaced textual lists with bar graphs to support faster and more intuitive processing. Beyond these conventional nutrition labeling strategies, we introduced additional adaptations specifically tailored to the needs of older adults, drawing on findings from prior research in aging-focused visualization design [2], [8], [12]. For example, the interface adopts a relatively large base font size (≥ 14 pt) to enhance legibility, reducing the strain of read-

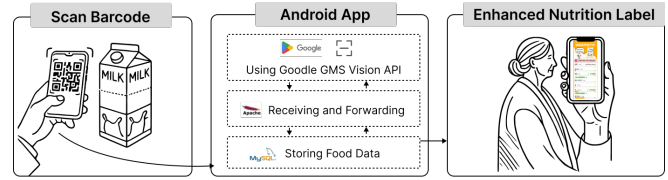


Fig. 3: Application Usage Flow

ing dense information and accommodating the age-related decline in visual acuity. In addition, to mitigate the slower information-processing speed often observed in older adults [10], we streamlined the interface using visual techniques. Specifically, we replaced lengthy textual descriptions with clear, intuitive icons in the ② Health Alerts and ⑤ Nutrient Emoji Tags. We also incorporated playful cartoon graphics, such as a heart and bones, into the ⑥ Daily Recommended Progress Bar to foster engagement and aid memory recall. Similarly, the ① Product Total Intake feature was designed as a speech bubble, with a talking character tailored to each product. For instance, in the case of milk (as shown in Figure 2), the message is delivered by a cartoon calf. These emotionally engaging and familiar visuals are intended to capture attention and support recall, particularly among older adults.

B. Technical Design Framework

Our food data is managed using MySQL and sourced from professional databases, including Open Food Facts, the US Food Database, and the “Dietary Nutrient Reference Intakes for Chinese Residents,” ensuring accuracy and scientific validity. The system workflow (Figure 3) begins with users scanning a food item’s barcode via an Android app using Google GMS Vision. The barcode data is sent to the server through WebView, where the Apache web server processes the request. PHP queries the integrated databases, and the results are displayed to the user through a responsive visualization interface designed with Bootstrap.

V. EXPERIMENT

The primary objective of this study is to evaluate whether the enhanced nutrition label system can improve elderly users’ understanding of food nutrition information and support more informed food choices. To achieve this, we conducted a real-world experiment comparing traditional and enhanced nutrition labels. We selected four frequently purchased food categories—milk, cereal powder, pastries, and ham—based on their relevance to elderly consumers and diverse nutritional profiles: Milk and cereal powder represent staple foods with essential nutrients, suitable for testing the enhanced labels’ ability to promote healthy dietary choices. Pastries and ham are foods high in sugar and fat, chosen to evaluate the labels’ effectiveness in highlighting potential health risks. This selection ensures the study covers a broad range of

dietary scenarios, providing a comprehensive and representative assessment of the enhanced labels.

A. Participants and Apparatus

This study involved 24 elderly participants (aged 50 and older, $M = 62.54$, $SD = 7.75$) recruited from local gathering places including squares and markets. The sample was gender-balanced (11 males, 13 females) and representative of diverse educational backgrounds: 25% had primary education, 29.2% each had junior high or high school education, and 16.7% had college education. Most participants (87.5%) were retired, and 23 had common chronic conditions. Participants were randomly assigned to four experimental groups (6 per group) with counterbalanced food evaluation orders to minimize sequence effects. For instance, Group 1 evaluated foods in the order of milk, cereal powder, pastries, and ham, while Group 2 followed cereal powder, pastries, ham, and milk. The experiment was conducted using a Google Pixel 5 smartphone with a 6-inch OLED screen (2340×1080 pixels), ensuring clear and accurate visual presentation.

B. Procedure

This study was approved by the university's ethics committee. Participants were provided with detailed information about the study's purpose, procedure, and their rights, and informed consent was obtained. The experiment lasted approximately 62 minutes and included the following phases: 1) **Background Questionnaire**: Participants completed a questionnaire to collect demographic details and relevant background information; 2) **Initial Evaluation**: Using traditional nutrition labels, participants evaluated selected foods and completed a questionnaire on their comprehension of the labels and their food choices; 3) **Enhanced Label Evaluation**: Participants used a mobile app to scan QR codes on the food items and evaluate the same set of foods with enhanced nutrition labels. After scanning the products in each category, they completed a questionnaire focusing on label comprehension, food choices, user experience, and satisfaction; 4) **Final Interview**: A 5-10 minute interview was conducted to gather additional feedback on the labels.

VI. RESULTS

We collected questionnaire data to examine changes in user behavior and nutrition knowledge before and after using our new labels. In the questionnaire, each correct answer was coded as 1, while incorrect answers and "I don't know" were coded as 0 to facilitate quantitative analysis. Since the same set of participants completed the questionnaire both before and after using the new labels/system, we conducted a paired-sample t -test on their knowledge scores (as well as on product selection accuracy) to evaluate any significant pre-post differences. In addition to reporting t and p values, we also report Cohen's d to assess the practical significance of the observed differences. For each design feature, participants

rated their satisfaction on a 5-point scale (from "strongly disagree" to "strongly agree"). We conducted descriptive analyses of these ratings and visualized their distribution via stacked bar charts to provide further insight.

A. Nutritional Product Selection Evaluation

For the nutritional product selection evaluation, after viewing either the traditional or enhanced labels on four categories of food, participants answered three questions about each food category, resulting in a total of 12 questions in the questionnaire (see Table II). Through these responses, we aimed to examine older adults' understanding of health products and to assess their overall comprehension. The data analysis results are presented in Figure 5). A paired-sample comparison (traditional labels vs. enhanced labels) revealed a significant difference in participants' nutritional product selection accuracy. Overall, the enhanced label condition yielded substantially higher accuracy compared with the traditional label condition. Before the experiment, the average accuracy was 39.9% ($SD = 0.128$), whereas after the experiment, it rose to 64.9% ($SD = 0.147$)—a clear improvement associated with the enhanced labels. A paired-samples t -test showed $t(23) = -6.281295$, $p < 0.0001$, Cohen's $d = 1.81$ (substantially greater than 0.8), indicating a very large effect size. When analyzing each food category separately (see Table IV), the enhanced labels consistently outperformed the traditional labels. All p -values were below 0.05, confirming statistically significant improvements in selection accuracy across all four categories.

B. Nutritional Knowledge Assessment

To evaluate nutritional knowledge awareness, six specialized questions were designed to assess whether the elderly participants' understanding of nutrition improved after using enhanced labels. These questions focused on common nutrients: calcium, protein, fat, sodium, dietary fiber, and iron, as detailed in Table III. Participants answered each question before and after the intervention, resulting in a total of 12 responses per participant. The data analysis results are presented in Figure 5. A paired-sample comparison (traditional labels vs. enhanced labels) revealed a significant improvement in participants' overall nutritional knowledge accuracy. The average accuracy rate increased from 41% ($SD = 0.166$) before the experiment to 64% ($SD = 0.138$) after the experiment, indicating a substantial enhancement associated with the use of enhanced labels. A paired-samples t -test showed $t(23) = -3.052$, $p = 0.003$, and Cohen's $d = 0.89$, indicating a large effect size. These results suggest that the enhanced labels had a significant impact on improving the nutritional knowledge of elderly participants. All six nutritional knowledge questions (Q1 to Q6) demonstrated significant improvements in accuracy after the intervention with enhanced labels (see Table V).

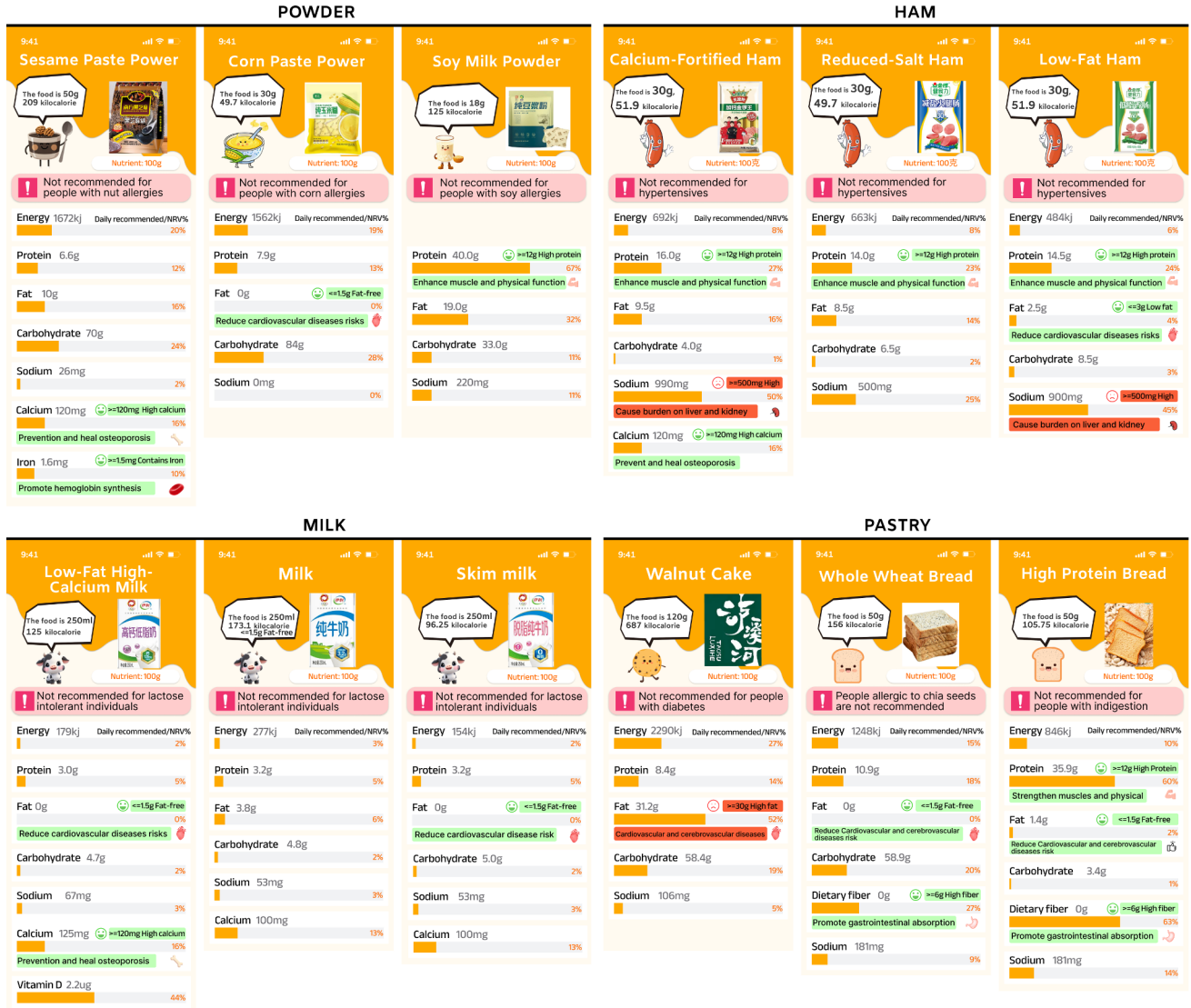


Fig. 4: Collage of Nutrition Labels for 4 Product Categories, Each with 3 Products

C. User Experience & Design Feature Evaluation

The overall label evaluation is presented in Figure 6, where six dimensions were scored and analyzed. In general, the labels we designed received high scores across all six dimensions in the functional evaluation task. Specifically, more than 75% of participants rated each dimension as 4 (Agree) or 5 (Strongly Agree), indicating a very high level of satisfaction and recognition of the labels in each dimension.

1) Evaluation on User Experience

The results indicate that all six dimensions achieved above-average scores, highlighting strong user satisfaction with the enhanced labels. Among these dimensions, **Readability** scored the highest, reflecting excellent performance in text clarity and layout. On the other hand, **Practicality**, while

still above average, received the lowest score. This suggests that, despite positive perceptions of the labels across other dimensions, participants may have encountered challenges in applying the enhanced labels to real-life nutritional decision-making for diverse products. Future efforts should focus on refining the design through a more detailed *co-design* process, gathering specific feedback to better align the labels with practical everyday use cases. Table VI presents the mean scores and standard deviations for each dimension and Figure 6 presents distribution for each.

2) Design Feature Preferences

The preferences of elderly participants for seven design features of the enhanced labels were collected using a 5-point satisfaction scale (1 = Most Disagree, 5 = Strongly

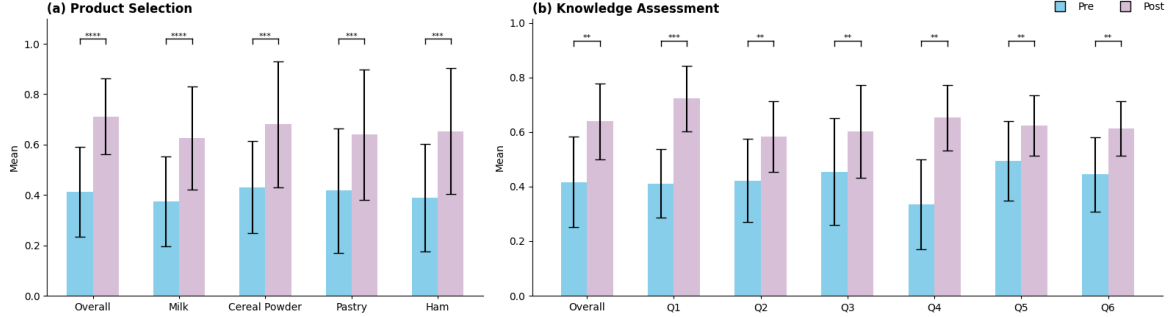


Fig. 5: Average accuracy on product selection and nutrition knowledge assessment. Significance levels: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), and $p < 0.0001$ (****).

Agree). The results, summarized in Table VII, indicate an overall positive response to the enhanced labels, with six modules achieving over 80% positive ratings. The mean scores ranged from 3.88 to 4.29, demonstrating consistent satisfaction across features. Several key patterns emerge from the data:

1. Highly Favored Features: **Nutrient Emoji Tags** ($M = 4.29$) was the most favored feature, suggesting that the use of emojis enhances the fun and accessibility of nutritional information, making it easier for elderly users to understand. **Health Alerts** ($M = 4.25$) ranked second, indicating that participants valued critical health-related alerts.
2. Moderately Strong Features: **Nutrient Thresholds** and **Nutrient Interpretation** both scored 4.17, demonstrating their effectiveness in communicating specific nutrient details. However, their performance suggests opportunities for subtle refinements to improve user experience further. **Nutrient Text Tags** ($M = 4.12$) performed well, reflecting good acceptance of text-based information presentation, but it was not as engaging as other features like emoji tags.
3. Features Needing Optimization: **Product Total** ($M = 4.00$) received a relatively lower score, indicating that while participants found it somewhat helpful, its role in decision-making could be further clarified. **Daily Recommended Progress Bar** ($M = 3.88$) scored the lowest, highlighting potential usability issues.

VII. DISCUSSION

Reflection on Label Design for Elderly. Our experiment revealed that elderly users prioritize information related to interpreting nutrient values and their relevance to specific health conditions. When presented with nutrient data, they overwhelmingly preferred simple, recognizable icons (e.g., smiley or sad faces) to convey the meaning of values—a finding consistent with prior research [23], [8] on elderly-friendly visualization. From the outset, our design process was informed by extensive user research and prior studies on visualization for elderly populations. This foundation contributed to the high overall satisfaction with the final design.

Additionally, recognizing the elderly's limited familiarity with AR technology, we opted for an Android application to deliver the content directly on devices they are more accustomed to using. This approach ensured accessibility while maintaining technological feasibility. Overall, our design process emphasized user-centered principles, drawing insights from existing literature and tailored user feedback to create a solution that balances simplicity, familiarity, and effectiveness for elderly users.

Limitations and Future Work. While this study demonstrates the potential of enhanced nutrition labels for improving comprehension and decision-making among elderly users, several limitations remain. The findings are based on participants from economically developed urban areas, which may limit generalizability to rural or less affluent populations. Additionally, the current design's comprehensive information may overwhelm some users, highlighting the need for future iterations to include personalization features that adapt to individual health needs and simplify information presentation. Lastly, the long-term impacts of these labels on dietary behavior and health outcomes are yet to be determined, warranting longitudinal studies across diverse socioeconomic and health contexts to validate their sustained effectiveness.

VIII. CONCLUSION

This study targeted elderly individuals, using questionnaires to examine their current use and comprehension of food labels, revealing significant difficulties and misunderstandings. Based on prior design guidelines for elderly users, we developed an enhanced nutrition label. Participants evaluated four food categories (each with three similar products) using both traditional and newly designed labels. We compared their nutrition knowledge and product selection accuracy before and after using the new labels and gathered usability feedback and ratings for each design element. Results showed that the enhanced labels outperformed traditional ones across all metrics, with over 75% of users giving positive or moderately positive feedback on design features. These findings provide

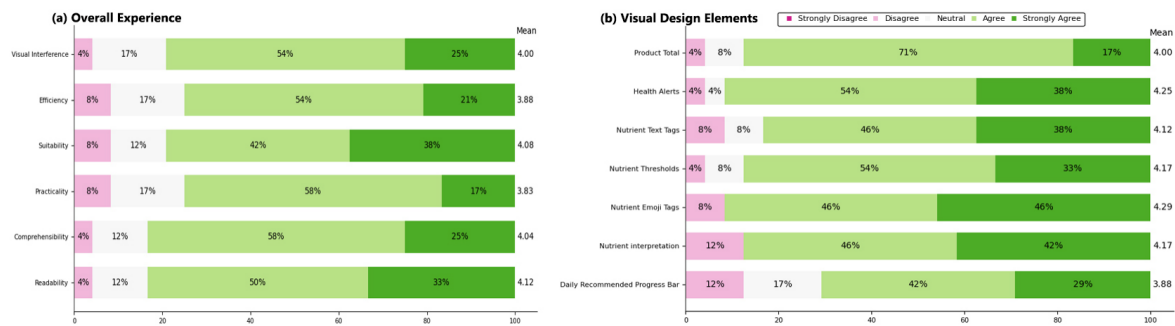


Fig. 6: Subjective ratings on user experience and each visual feature of the enhanced label.

valuable insights for improving nutrition labels tailored to elderly users.

ACKNOWLEDGMENT

The work was supported by XJTLU RDF(22-01-092).

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